

# Epigenetic Control of Temperature Response in *Ostreococcus tauri* (ECTOR)

Christina Arvanitidou (carvanitidou@us.es) and Francisco J. Romero-Campero (fran@us.es)

2024-03-15

## H3K27 genome wide distribution analysis

We load the genome of *Ostreococcus tauri* and compute the length of its chromosomes. We determine the length of the largest chromosome and for technical reasons we reverse the vector of lengths. This will be used to represent graphically the genome wide distribution of H3K27me3.

```
library(seqinr)

ostta.genome <- read.fasta(file = "ostta_genome/ostreococcus_tauri.fa", seqtype = "DNA")
chr.lens <- sapply(X = getSequence(ostta.genome), FUN = length)

number.chrs <- length(getSequence(ostta.genome))
max.chr.len <- max(chr.lens)
chr.lens <- rev(chr.lens)
```

We load the H3K27me3 peaks identified in cultures grown under constant light at 20°C. We represent each chromosome using a rectangle. Each H3K27me3 peak is represented in its precise genome location using a blue rectangle.

```
ll.20 <- read.table(file="chipseq_data/bed_data/h3k27me3/h3k27me3_ll_20.bed",
                    header = F)

ll.20 <- ll.20[,1:3]

nrow(ll.20)

## [1] 480

#jpeg(filename = "images/h3k27me3_genome_wide_distribution_ll_20.jpg", res = 300, width = 900, height = 900)
#par(mar=c(0,0,0,0))
plot(x=c(0,max.chr.len), y=c(0,4*number.chrs), col = "white", xlab = "", ylab = "", axes=F)

for(i in 1:number.chrs)
{
  polygon(x = c(0, chr.lens[i], chr.lens[i], 0),
          y = c(4*i+1, 4*i+1, 4*i-1, 4*i-1),
          col = "white")
}

for(i in 1:nrow(ll.20))
{
  current.chr <- ll.20[i,1]
```

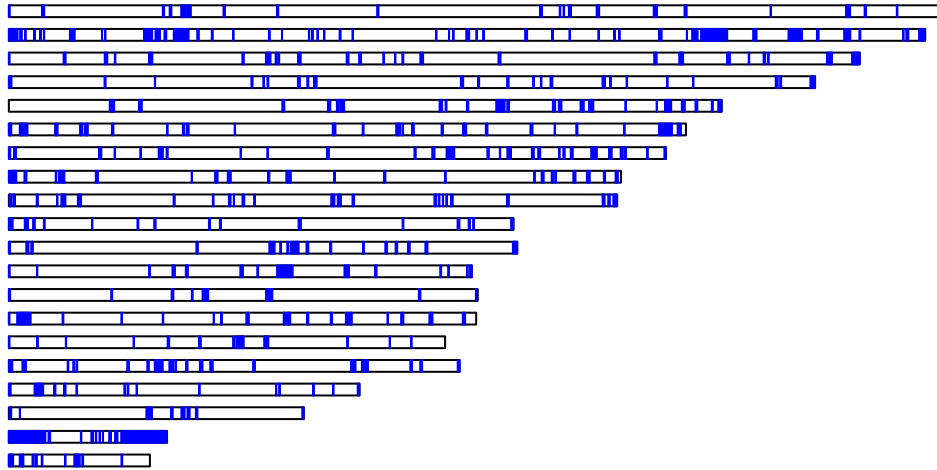
```

current.start <- ll.20[i,2]
current.end <- ll.20[i,3]

current.line <- (20:1)[current.chr]

polygon(x = c(current.start, current.end, current.end, current.start),
        y = c(4*current.line+1, 4*current.line+1, 4*current.line-1, 4*current.line-1),
        col = "blue",border="blue")
}

```



```
#dev.off()
```

We identified 480 genomic regions significantly occupied by H3K27me3 in the three different independent biological replicates. We compute the percentage of the whole genome occupied by H3K27me3 and the percentage in each chromosome. A barplot is used to represent the percentage of H3K27me3 in each chromosome. Z-scores and p-values are computed for each chromosome. A gradient from white to blue is used to represent significance levels.

```

h3k27me3.peaks.len <- ll.20$V3 - ll.20$V2
sum(h3k27me3.peaks.len)/sum(chr.lens)

```

```
## [1] 0.07163182
```

```

ll.20 <- cbind(ll.20,h3k27me3.peaks.len)
colnames(ll.20) <- c("chr","start","end","length")
head(ll.20)

```

```

##   chr  start    end length
## 1   1    174   1061    887
## 2   1 38825 41425   2600
## 3   1 180446 180824    378
## 4   1 181798 181979    181
## 5   1 188479 190520   2041
## 6   1 202039 207172   5133

```

```

number.of.h3k27.peaks.chr <- vector(mode = "numeric",length = 20)
peaks.len.chr <- vector(mode = "numeric",length = 20)

```

```

for(i in 1:20)
{
  number.of.h3k27.peaks.chr[i] <- nrow(subset(ll.20, chr == i))
}

```

```

peaks.len.chr[i] <- sum(subset(ll_20, chr == i)[,4])
}

percentage.chr.k27 <- peaks.len.chr/rev(chr.lens)

z.scores <- (percentage.chr.k27 - mean(percentage.chr.k27))/sd(percentage.chr.k27)

which(z.scores > 2.5)

## [1] 19

p.vals <- pnorm((percentage.chr.k27 - mean(percentage.chr.k27[-19]))/sd(percentage.chr.k27[-19]),lower.tail=FALSE)

p.vals

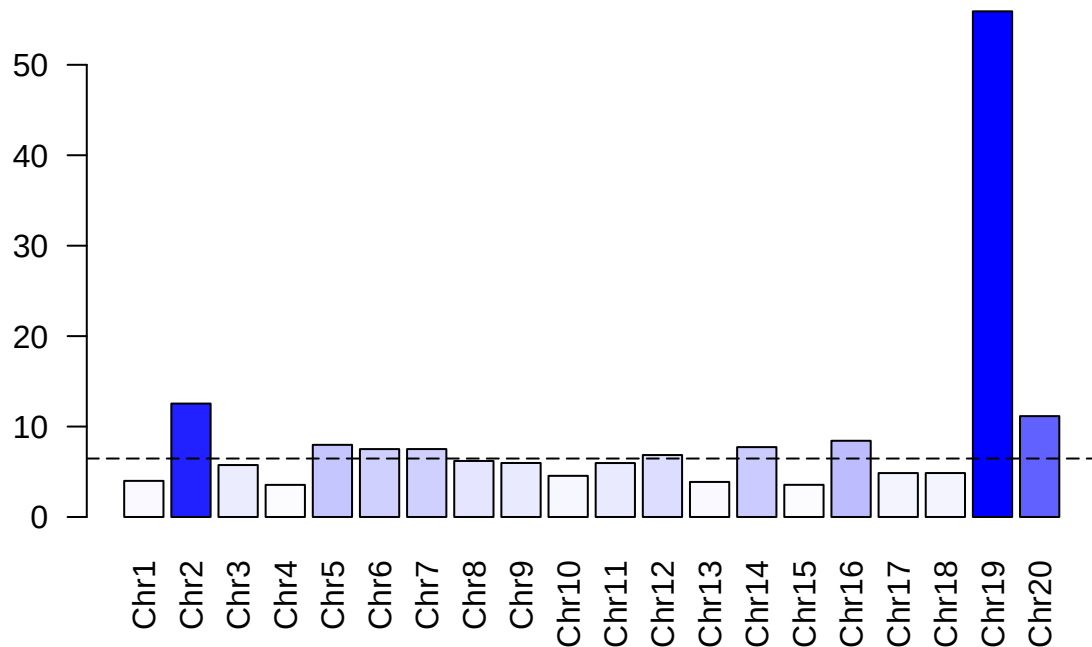
## [1] 8.428706e-01 6.683267e-03 6.150346e-01 8.821057e-01 2.690296e-01
## [6] 3.356068e-01 3.348786e-01 5.436769e-01 5.792470e-01 7.820867e-01
## [11] 5.803782e-01 4.374461e-01 8.540130e-01 3.039144e-01 8.815987e-01
## [16] 2.119113e-01 7.434354e-01 7.430300e-01 2.062177e-90 2.815232e-02

values.for.colors <- round(-log10(p.vals)*40)
values.for.colors[19] <- 100

white.blue.palette <- colorRampPalette(c('white','blue'))

#jpeg(filename = "images/h3k27me3_chr_barplot_ll_20.jpg",res = 300,width = 1800,height = 900)
barplot(100*percentage.chr.k27,border="black",names.arg = paste0("Chr",1:20),las=2,col=white.blue.palette[values.for.colors])
abline(h = 100*mean(percentage.chr.k27[-19]),lty=5)

```



```
#dev.off()
```

Approximately 7.2% of the *Ostreococcus* genome is occupied by H3K27me3. H3K27me3 is mostly uniformly distributed over the different chromosome although it significantly accumulates (p-value < 0.01) in Chr2 (sexual chromosome) and Chr19 (viral resistance chromosome) with p-values  $6.68 \times 10^{-3}$  and  $2.06 \times 10^{-90}$  respectively.

We determine the gene features occupied by each H3K27me3 mark and the H3K27me3 marked genes.

```
library(TxDb.Otauri.JGI)
```

```
## Loading required package: GenomicFeatures
## Loading required package: BiocGenerics
##
## Attaching package: 'BiocGenerics'
## The following objects are masked from 'package:stats':
##
##     IQR, mad, sd, var, xtabs
## The following objects are masked from 'package:base':
##
##     anyDuplicated, append, as.data.frame, basename, cbind, colnames,
##     dirname, do.call, duplicated, eval, evalq, Filter, Find, get, grep,
##     grepl, intersect, is.unsorted, lapply, Map, mapply, match, mget,
##     order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
##     rbind, Reduce, rownames, sapply, setdiff, sort, table, tapply,
##     union, unique, unsplit, which.max, which.min
## Loading required package: S4Vectors
## Loading required package: stats4
##
## Attaching package: 'S4Vectors'
## The following objects are masked from 'package:base':
##
##     expand.grid, I, unname
## Loading required package: IRanges
## Loading required package: GenomeInfoDb
## Loading required package: GenomicRanges
## Loading required package: AnnotationDbi
## Loading required package: Biobase
## Welcome to Bioconductor
##
##     Vignettes contain introductory material; view with
##     'browseVignettes()'. To cite Bioconductor, see
##     'citation("Biobase")', and for packages 'citation("pkgname)".
txdb <- TxDb.Otauri.JGI

ostta.genes <- as.data.frame(genes(txdb))
head(ostta.genes)
```

```
##           seqnames start   end width strand   gene_id
## ostta01g00010      1    244  1932  1689      - ostta01g00010
## ostta01g00020      1   2077   4029  1953      + ostta01g00020
## ostta01g00030      1   4157   5299  1143      + ostta01g00030
## ostta01g00040      1   5392   7041  1650      - ostta01g00040
## ostta01g00050      1   7386   9605  2220      - ostta01g00050
```

```

## ostta01g00060      1  9780 10412   633      - ostta01g00060
target.genes <- vector(mode = "character",length = nrow(ll.20))
peak.annotation <- vector(mode = "character",length = nrow(ll.20))

for(i in 1:nrow(ll.20))
{
  current.chr <- ll.20[i,1]
  current.start <- ll.20[i,2]
  current.end <- ll.20[i,3]

  complete.gene.body <- subset(ostta.genes,
                              seqnames == current.chr &
                              start >= current.start &
                              end <= current.end)

  overlap.tss.pos.strand <- subset(ostta.genes,
                                   seqnames == current.chr &
                                   start >= current.start & start <= current.end &
                                   strand == "+")

  overlap.tss.neg.strand <- subset(ostta.genes,
                                   seqnames == current.chr &
                                   end >= current.start & end <= current.end &
                                   strand == "-")

  overlap.tes.pos.strand <- subset(ostta.genes,
                                   seqnames == current.chr &
                                   end >= current.start & end <= current.end &
                                   strand == "+")

  overlap.tes.neg.strand <- subset(ostta.genes,
                                   seqnames == current.chr &
                                   start >= current.start & start <= current.end &
                                   strand == "-")

  inside.gene.body <- subset(ostta.genes, seqnames == current.chr &
                             start <= current.start & end >= current.end)

  if(nrow(complete.gene.body) > 0)
  {
    target.genes[i] <- paste(unlist(complete.gene.body$gene_id),collapse = ",")
    peak.annotation[i] <- "gene.body"
  } else if(nrow(overlap.tss.pos.strand) > 0)
  {
    target.genes[i] <- paste(unlist(overlap.tss.pos.strand$gene_id),collapse = ",")
    peak.annotation[i] <- "TSS"
  } else if(nrow(overlap.tss.neg.strand) > 0)
  {
    target.genes[i] <- paste(unlist(overlap.tss.neg.strand$gene_id),collapse = ",")
    peak.annotation[i] <- "TSS"
  } else if(nrow(overlap.tes.pos.strand) > 0)
  {

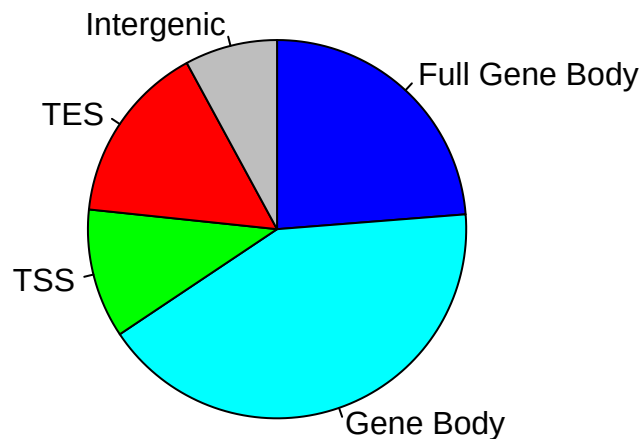
```

```

target.genes[i] <- paste(unlist(overlap.tes.pos.strand$gene_id),collapse = ",")
peak.annotation[i] <- "TES"
} else if(nrow(overlap.tes.neg.strand) > 0)
{
  target.genes[i] <- paste(unlist(overlap.tes.neg.strand$gene_id),collapse = ",")
  peak.annotation[i] <- "TES"
} else if(nrow(inside.gene.body) > 0)
{
  target.genes[i] <- paste(unlist(inside.gene.body$gene_id),collapse = ",")
  peak.annotation[i] <- "internal.gene.body"
} else
{
  peak.annotation[i] <- "intergenic"
}
}

freq.peak.annotation <- table(peak.annotation)
#jpeg(filename = "images/h3k27me3_pie_chart_ll_20.jpg",res = 300,width = 900,height = 900)
#par(mar=c(0,0,0,0))
pie(freq.peak.annotation[c("gene.body",
                           "internal.gene.body",
                           "TSS",
                           "TES",
                           "intergenic")],
    labels = c("Full Gene Body",
               "Gene Body",
               "TSS",
               "TES",
               "Intergenic"),
    clockwise = T, border = "black", col=c("blue","cyan","green","red","grey"))

```



```

#dev.off()

percentage.freq.peak.annotation <- 100*freq.peak.annotation/sum(freq.peak.annotation)

percentage.freq.peak.annotation["gene.body"] +
percentage.freq.peak.annotation["internal.gene.body"]

## gene.body

```

```
##      65.625
percentage.freq.peak.annotation

## peak.annotation
##      gene.body      intergenic internal.gene.body      TES
##      23.750000      7.916667      41.875000      15.416667
##      TSS
##      11.041667

ll.20.peak.annotation <- cbind(ll.20[,1:3],data.frame(peak.annotation,target.genes))

head(ll.20.peak.annotation)

##   chr  start    end  peak.annotation target.genes
## 1   1    174   1061          TES ostta01g00010
## 2   1   3825  41425      gene.body ostta01g00260
## 3   1 180446 180824      intergenic
## 4   1 181798 181979      intergenic
## 5   1 188479 190520 internal.gene.body ostta01g01270
## 6   1 202039 207172          TSS ostta01g01340

write.table(x = ll.20.peak.annotation,file = "tables/H3K27me3_LL_20C_peak_annotation.tsv",quote = F,sep

all.k27.genes <- (unlist(sapply(X =ll.20.peak.annotation$target.genes,
                               FUN = function(x) {strsplit(x,",")[[1]]})))
names(all.k27.genes) <- NULL
length(all.k27.genes)

## [1] 491

length(all.k27.genes)/7668

## [1] 0.06403234
```

We found 491 genes associated to H3K27me3 marks corresponding to 6.4% of the entire *Ostreococcus* genome. H3K27me3 is mostly located occupying either the entire gene body (23.75%) or internal regions of gene body (41.86%). Although some regions occupied by H3K27me3 has been found overlapping only with the TSS (11.04%) or the TES (15.42%). H3K27me3 was found in intergenic regions only in 7.92% of the cases.

Next we use metageneplots to determine the shape of the H3K27me3 over its occupied genes. We consider three different sets of marked genes, genes with their entire gene body marked, genes with H3K27 inside their gene body with no overlap in the TSS or TES and genes with H3K27me3 overlapping their TSS or TES

```
## Function to compute ChIP signal over different tiles
ntile.signal <- function(signal.vector,number.tiles)
{
  step.len <- floor(length(signal.vector)/number.tiles)
  number.plus.1 <- length(signal.vector) %% number.tiles
  pos.plus.1 <- sort(sample(1:100,number.plus.1,replace = F))

  acum <- c(0)
  for(i in 1:number.tiles)
  {
    if(i %in% pos.plus.1)
    {
      acum <- c(acum,acum[i]+step.len+1)
    } else
  }
```

```

{
  acum <- c(acum,acum[i]+step.len)
}
}

acum[1] <- 1

n.tile.signal.vector <- vector(mode = "numeric",length = number.tiles)
for(i in 2:length(acum))
{
  n.tile.signal.vector[(i-1)] <- mean(signal.vector[acum[i-1]:acum[i]])
}

return(n.tile.signal.vector)
}

## Function to generate the data for a metageneplot
metageneplot <- function(gene.names,genes,bw.file.name,ntile.txs,ntile.gene.body)
{
  res <- matrix(nrow=length(gene.names),ncol=2*ntile.txs+ntile.gene.body)
  rownames(res) <- gene.names

  bw.file <- import.bw(con = bw.file.name)
  bw.file <- as.data.frame(bw.file)

  for(i in 1:length(gene.names))
  {
    #print(i)
    current.gene.chr <- genes[gene.names[i],"seqnames"]
    current.gene.start <- genes[gene.names[i],"start"]
    current.gene.end <- genes[gene.names[i],"end"]
    current.gene.strand <- genes[gene.names[i],"strand"]

    gene.body.signal <- subset(bw.file, seqnames == current.gene.chr &
                             start >= current.gene.start &
                             end <= current.gene.end)[["score"]]

    upstream.start.signal <- subset(bw.file, seqnames == current.gene.chr &
                                    start >= (current.gene.start - 2000) &
                                    end <= current.gene.start)[["score"]]

    downstream.end.signal <- subset(bw.file, seqnames == current.gene.chr &
                                    start >= current.gene.end &
                                    end <= (current.gene.end + 2000))[["score"]]

    ntile.signal.complete <-
      c(ntile.signal(signal.vector = upstream.start.signal,number.tiles = ntile.txs),
        ntile.signal(signal.vector = gene.body.signal,number.tiles = ntile.gene.body),
        ntile.signal(signal.vector = downstream.end.signal,number.tiles = ntile.txs))

    if(current.gene.strand == "+")
    {
      res[i,] <- ntile.signal.complete
    }
  }
}

```

```

} else
{
  res[i,] <- rev(ntile.signal.complete)
}
}

return(res)
}

## Genes with their entire gene body marked
k27.gene.body <- (unlist(sapply(X = subset(11.20.peak.annotation, peak.annotation == "gene.body")["target.genes"],
  FUN = function(x) {strsplit(x, ",")[[1]]})))
names(k27.gene.body) <- NULL
length(k27.gene.body)

## [1] 160

## Genes with H3K27 inside their gene body with no overlap
## in the TSS or TES.
k27.internal.gene.body <- (unlist(sapply(X = subset(11.20.peak.annotation, peak.annotation == "internal.gene.body")["target.genes"],
  FUN = function(x) {strsplit(x, ",")[[1]]})))
names(k27.internal.gene.body) <- NULL
length(k27.internal.gene.body)

## [1] 203

## Genes with H3K27me3 overlapping their TSS
k27.tss <- (unlist(sapply(X = subset(11.20.peak.annotation, peak.annotation == "TSS")["target.genes"],
  FUN = function(x) {strsplit(x, ",")[[1]]})))
names(k27.tss) <- NULL
length(k27.tss)

## [1] 54

## Genes with H3K27me3 overlapping their TES
k27.tes <- (unlist(sapply(X = subset(11.20.peak.annotation, peak.annotation == "TES")["target.genes"],
  FUN = function(x) {strsplit(x, ",")[[1]]})))
names(k27.tes) <- NULL
length(k27.tes)

## [1] 74

```

Next we generate the metagenepLOTS for genes where the H3K27me3 occupies their entire gene body including TSS and TES.

```

library(rtracklayer)

## Genes with their entire gene body marked
k27.gene.body <- (unlist(sapply(X = subset(11.20.peak.annotation, peak.annotation == "gene.body")["target.genes"],
  FUN = function(x) {strsplit(x, ",")[[1]]})))
names(k27.gene.body) <- NULL
length(k27.gene.body)

## [1] 160

## Genes with H3K27 inside their gene body with no overlap
## in the TSS or TES.
k27.internal.gene.body <- (unlist(sapply(X = subset(11.20.peak.annotation, peak.annotation == "internal.gene.body")["target.genes"],
  FUN = function(x) {strsplit(x, ",")[[1]]})))

```

```
names(k27.internal.gene.body) <- NULL
length(k27.internal.gene.body)
```

```
## [1] 203
```

```
## Genes with H3K27me3 overlapping their TSS
```

```
k27.tss <- (unlist(sapply(X = subset(11.20.peak.annotation, peak.annotation == "TSS")["target.genes"],
                           FUN = function(x) {strsplit(x, ",")[[1]]})))
```

```
names(k27.tss) <- NULL
length(k27.tss)
```

```
## [1] 54
```

```
## Genes with H3K27me3 overlapping their TES
```

```
k27.tes <- (unlist(sapply(X = subset(11.20.peak.annotation, peak.annotation == "TES")["target.genes"],
                           FUN = function(x) {strsplit(x, ",")[[1]]})))
```

```
names(k27.tes) <- NULL
length(k27.tes)
```

```
## [1] 74
```

```
metagenepplot.data.k27.gene.body.1 <-
```

```
  metagenepplot(gene.names = k27.gene.body,
                genes = ostta.genes,
                bw.file.name = "chipseq_data/bw_data/h3k27me3/temp20/h3k27me3_20_chip_1.bw",
                ntile.txs = 10,
                ntile.gene.body = 20)
```

```
median.metagenepplot.data.k27.gene.body.1 <-
```

```
  apply(X = metagenepplot.data.k27.gene.body.1,
        MARGIN = 2,
        FUN = median, na.rm = T)
```

```
metagenepplot.data.k27.gene.body.2 <-
```

```
  metagenepplot(gene.names = k27.gene.body,
                genes = ostta.genes,
                bw.file.name = "chipseq_data/bw_data/h3k27me3/temp20/h3k27me3_20_chip_2n.bw",
                ntile.txs = 10,
                ntile.gene.body = 20)
```

```
median.metagenepplot.data.k27.gene.body.2 <-
```

```
  apply(X = metagenepplot.data.k27.gene.body.2,
        MARGIN = 2,
        FUN = median, na.rm = T)
```

```
metagenepplot.data.k27.gene.body.3 <-
```

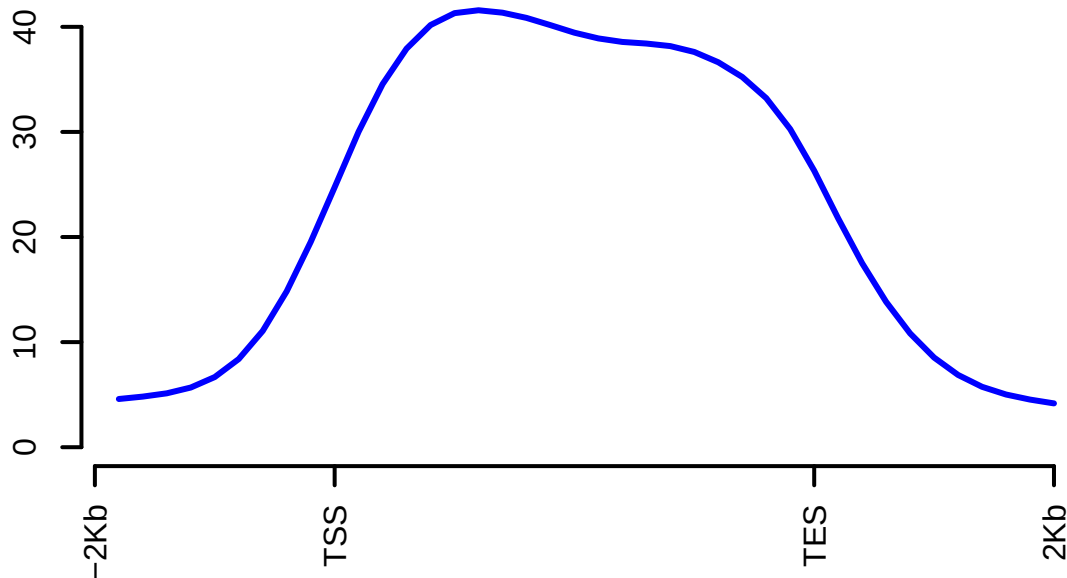
```
  metagenepplot(gene.names = k27.gene.body,
                genes = ostta.genes,
                bw.file.name = "chipseq_data/bw_data/h3k27me3/temp20/h3k27me3_20_chip_3n.bw",
                ntile.txs = 10,
                ntile.gene.body = 20)
```

```
median.metagenepplot.data.k27.gene.body.3 <-
```

```
  apply(X = metagenepplot.data.k27.gene.body.3,
        MARGIN = 2,
        FUN = median, na.rm = T)
```

```
median.metagenepplot.data.k27.gene.body <- (median.metagenepplot.data.k27.gene.body.1 + median.metagenepplot.data.k27.gene.body.2 + median.metagenepplot.data.k27.gene.body.3)
```

```
plot(smooth.spline(x=1:40,y=median.metageneplot.data.k27.gene.body,spar = 0.5),
     type="l",col="blue",lwd=3,ylim=c(0,45),axes=F,xlab="",ylab="")
axis(side = 1, at=c(0,10,30,40),labels = c("-2Kb","TSS","TES","2Kb"),las=2,lwd=2)
axis(side = 2,lwd=2)
```



Next we generate the metageneplots for genes where the H3K27me3 occupies part of the gene body without overlapping with the TSS and TES.

```
metageneplot.data.k27.internal.gene.body.1 <-
  metageneplot(gene.names = k27.internal.gene.body,
               genes = ostta.genes,
               bw.file.name = "chipseq_data/bw_data/h3k27me3/temp20/h3k27me3_20_chip_1.bw",
               ntile.txs = 10,
               ntile.gene.body = 20)
median.metageneplot.data.k27.internal.gene.body.1 <-
  apply(X = metageneplot.data.k27.internal.gene.body.1,
        MARGIN = 2,
        FUN = median,na.rm = T)

metageneplot.data.k27.internal.gene.body.2 <-
  metageneplot(gene.names = k27.internal.gene.body,
               genes = ostta.genes,
               bw.file.name = "chipseq_data/bw_data/h3k27me3/temp20/h3k27me3_20_chip_2n.bw",
               ntile.txs = 10,
               ntile.gene.body = 20)
median.metageneplot.data.k27.internal.gene.body.2 <-
  apply(X = metageneplot.data.k27.internal.gene.body.2,
        MARGIN = 2,
        FUN = median,na.rm = T)

metageneplot.data.k27.internal.gene.body.3 <-
  metageneplot(gene.names = k27.internal.gene.body,
               genes = ostta.genes,
               bw.file.name = "chipseq_data/bw_data/h3k27me3/temp20/h3k27me3_20_chip_3n.bw",
               ntile.txs = 10,
               ntile.gene.body = 20)
```

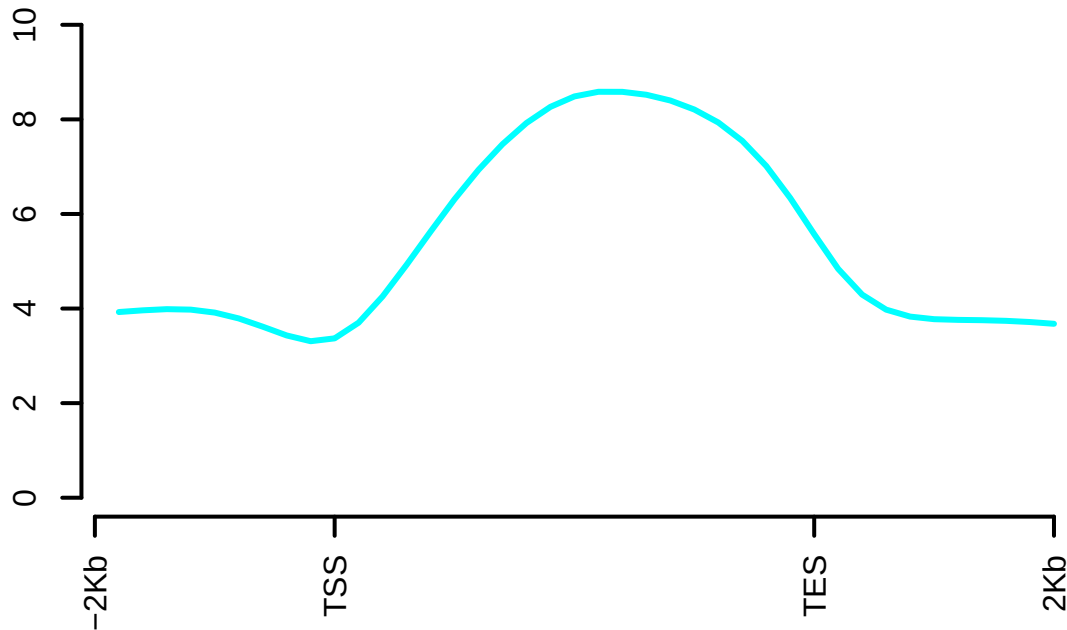
```

median.metagenesplot.data.k27.internal.gene.body.3 <-
  apply(X = metagenesplot.data.k27.internal.gene.body.3,
        MARGIN = 2,
        FUN = median, na.rm = T)

median.metagenesplot.data.k27.internal.gene.body <- (median.metagenesplot.data.k27.internal.gene.body.1 +
  median.metagenesplot.data.k27.internal.gene.body.2 + median.metagenesplot.data.k27.internal.gene.body.3)

plot(smooth.spline(x=1:40,y=median.metagenesplot.data.k27.internal.gene.body,spar = 0.5),
     type="l",col="cyan",lwd=3,ylim=c(0,10),axes=F,xlab="",ylab="")
axis(side = 1, at=c(0,10,30,40),labels = c("-2Kb","TSS","TES","2Kb"),las=2,lwd=2)
axis(side = 2,lwd=2)

```



Next we generate the metagenesplots for genes where the H3K27me3 occupies only the TSS and the beginning of the gene.

```

metagenesplot.data.k27.tss.1 <-
  metagenesplot(gene.names = k27.tss,
                genes = ostta.genes,
                bw.file.name = "chipseq_data/bw_data/h3k27me3/temp20/h3k27me3_20_chip_1.bw",
                ntile.txs = 10,
                ntile.gene.body = 20)
median.metagenesplot.data.k27.tss.1 <-
  apply(X = metagenesplot.data.k27.tss.1,
        MARGIN = 2,
        FUN = median, na.rm = T)

metagenesplot.data.k27.tss.2 <-
  metagenesplot(gene.names = k27.tss,
                genes = ostta.genes,
                bw.file.name = "chipseq_data/bw_data/h3k27me3/temp20/h3k27me3_20_chip_2n.bw",
                ntile.txs = 10,
                ntile.gene.body = 20)
median.metagenesplot.data.k27.tss.2 <-

```

```

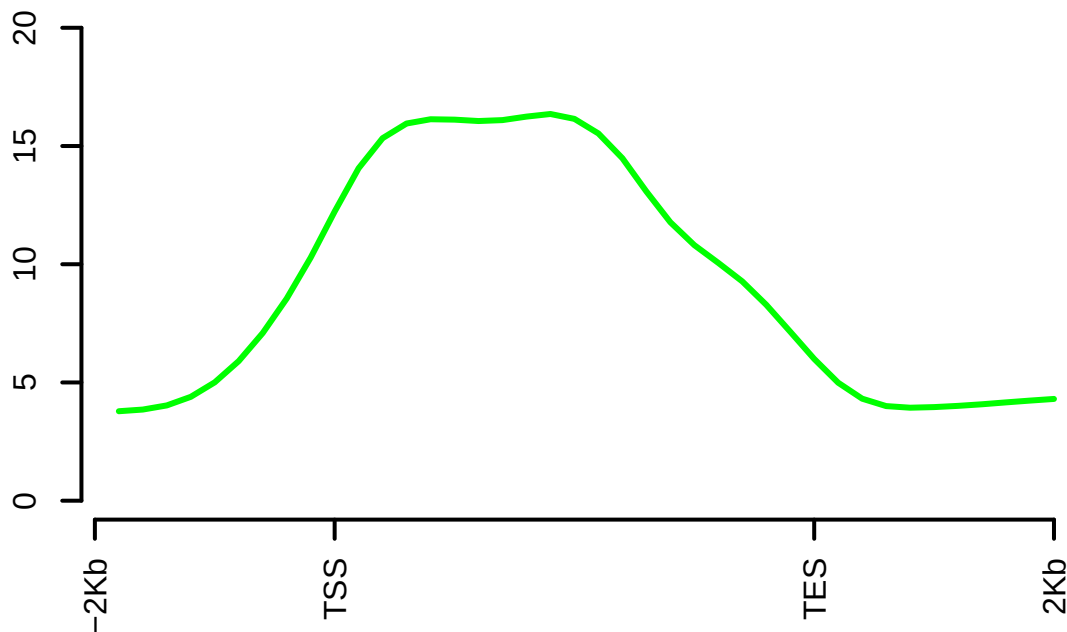
apply(X = metageneplot.data.k27.tss.2,
      MARGIN = 2,
      FUN = median, na.rm = T)

metageneplot.data.k27.tss.3 <-
  metageneplot(gene.names = k27.tss,
               genes = ostta.genes,
               bw.file.name = "chipseq_data/bw_data/h3k27me3/temp20/h3k27me3_20_chip_3n.bw",
               ntile.txs = 10,
               ntile.gene.body = 20)
median.metageneplot.data.k27.tss.3 <-
  apply(X = metageneplot.data.k27.tss.3,
        MARGIN = 2,
        FUN = median, na.rm = T)

median.metageneplot.data.k27.tss <-
  (median.metageneplot.data.k27.tss.1 +
   median.metageneplot.data.k27.tss.2 +
   median.metageneplot.data.k27.tss.3)

plot(smooth.spline(x=1:40,y=median.metageneplot.data.k27.tss,spar = 0.5),
     type="l",col="green",lwd=3,ylim=c(0,20),axes=F,xlab="",ylab="")
axis(side = 1, at=c(0,10,30,40),labels = c("-2Kb","TSS","TES","2Kb"),las=2,lwd=2)
axis(side = 2,lwd=2)

```



Next we generate the metageneplots for genes where the H3K27me3 occupies only the TSS and the beginning of the gene.

```

metageneplot.data.k27.tes.1 <-
  metageneplot(gene.names = k27.tes,
               genes = ostta.genes,
               bw.file.name = "chipseq_data/bw_data/h3k27me3/temp20/h3k27me3_20_chip_1.bw",
               ntile.txs = 10,

```

```

        ntile.gene.body = 20)
median.metagenepLOT.data.k27.tes.1 <-
  apply(X = metagenepLOT.data.k27.tes.1,
        MARGIN = 2,
        FUN = median,na.rm = T)

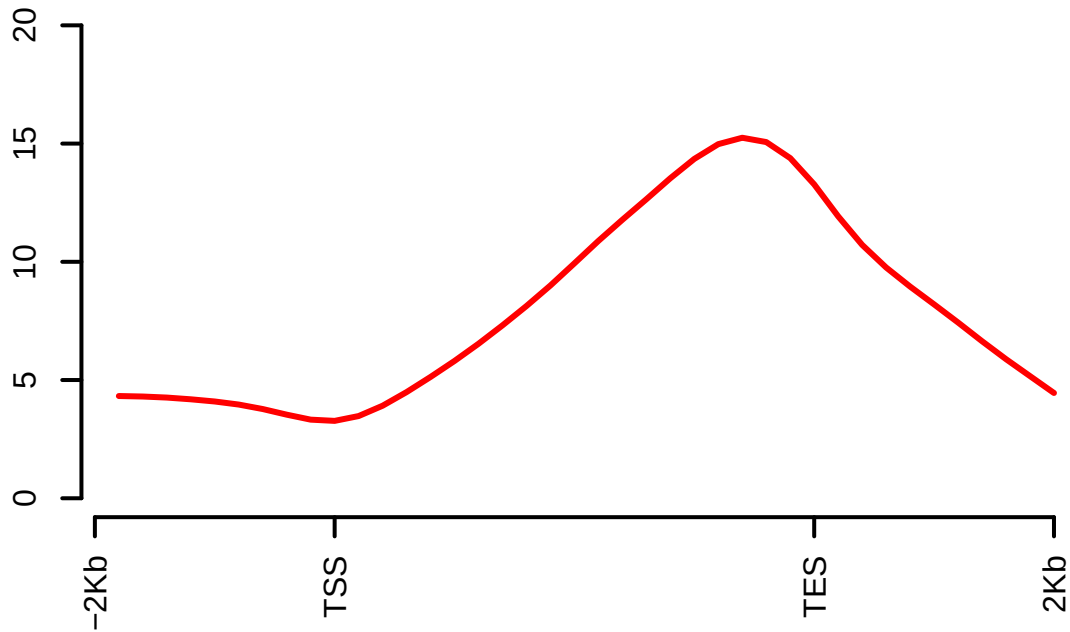
metagenepLOT.data.k27.tes.2 <-
  metagenepLOT(gene.names = k27.tes,
               genes = ostta.genes,
               bw.file.name = "chipseq_data/bw_data/h3k27me3/temp20/h3k27me3_20_chip_2n.bw",
               ntile.txs = 10,
               ntile.gene.body = 20)
median.metagenepLOT.data.k27.tes.2 <-
  apply(X = metagenepLOT.data.k27.tes.2,
        MARGIN = 2,
        FUN = median,na.rm = T)

metagenepLOT.data.k27.tes.3 <-
  metagenepLOT(gene.names = k27.tes,
               genes = ostta.genes,
               bw.file.name = "chipseq_data/bw_data/h3k27me3/temp20/h3k27me3_20_chip_3n.bw",
               ntile.txs = 10,
               ntile.gene.body = 20)
median.metagenepLOT.data.k27.tes.3 <-
  apply(X = metagenepLOT.data.k27.tes.3,
        MARGIN = 2,
        FUN = median,na.rm = T)

median.metagenepLOT.data.k27.tes <-
  (median.metagenepLOT.data.k27.tes.1 +
   median.metagenepLOT.data.k27.tes.2 +
   median.metagenepLOT.data.k27.tes.3)

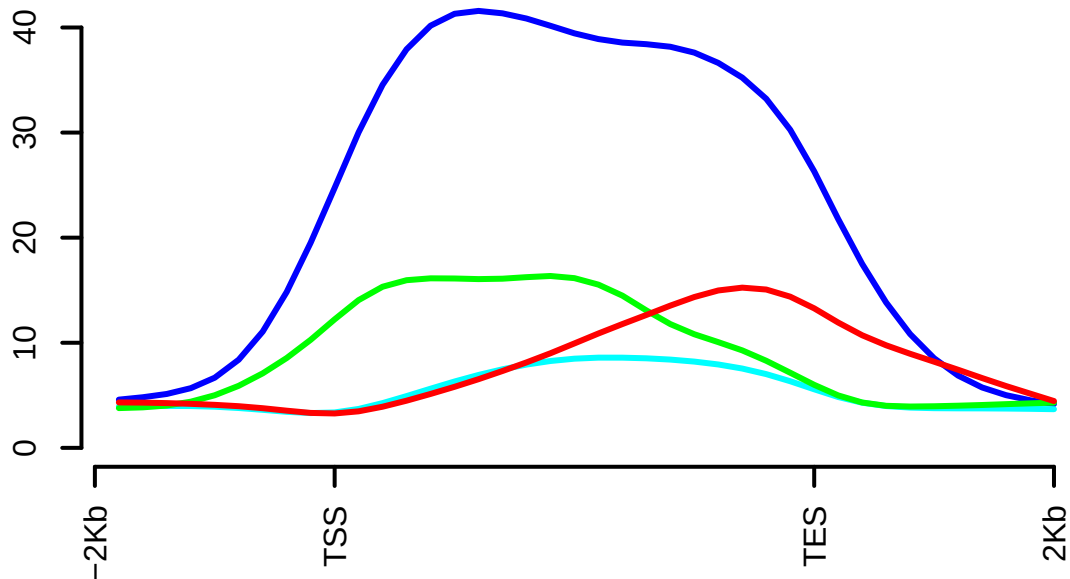
plot(smooth.spline(x=1:40,y=median.metagenepLOT.data.k27.tes,spar = 0.5),
     type="l",col="red",lwd=3,ylim=c(0,20),axes=F,xlab="",ylab="")
axis(side = 1, at=c(0,10,30,40),labels = c("-2Kb","TSS","TES","2Kb"),las=2,lwd=2)
axis(side = 2,lwd=2)

```



In order to compare the different metagenepoints are represented in the same plot.

```
#jpeg(filename = "images/h3k27me3_metagenepoints.jpg",res = 250,width = 900,height = 900)
plot(smooth.spline(x=1:40,y=median.metagenepoint.data.k27.gene.body,spar = 0.5),
     type="l",col="blue",lwd=3,ylim=c(0,45),axes=F,xlab="",ylab="")
lines(smooth.spline(x=1:40,y=median.metagenepoint.data.k27.internal.gene.body,spar = 0.5),
      type="l",col="cyan",lwd=3)
lines(smooth.spline(x=1:40,y=median.metagenepoint.data.k27.tss,spar = 0.5),
      type="l",col="green",lwd=3)
lines(smooth.spline(x=1:40,y=median.metagenepoint.data.k27.tes,spar = 0.5),
      type="l",col="red",lwd=3)
axis(side = 1, at=c(0,10,30,40),labels = c("-2Kb", "TSS", "TES", "2Kb"),las=2,lwd=2)
axis(side = 2,lwd=2)
```



```
#dev.off()
```

It can be observed that the level of H3K27me3 mark varies depending on the part of gene that it occupies. Next we check if these differences are significant.

```
max.metagenepplot.data.k27.gene.body.1 <-
  apply(X = metagenepplot.data.k27.gene.body.1,
        MARGIN = 1,
        FUN = max, na.rm = T)
max.metagenepplot.data.k27.gene.body.2 <-
  apply(X = metagenepplot.data.k27.gene.body.2,
        MARGIN = 1,
        FUN = max, na.rm = T)
max.metagenepplot.data.k27.gene.body.3 <-
  apply(X = metagenepplot.data.k27.gene.body.3,
        MARGIN = 1,
        FUN = max, na.rm = T)

max.metagenepplot.data.k27.gene.body <- (max.metagenepplot.data.k27.gene.body.1 +
                                          max.metagenepplot.data.k27.gene.body.2 +
                                          max.metagenepplot.data.k27.gene.body.3)/3

max.metagenepplot.data.k27.internal.gene.body.1 <-
  apply(X = metagenepplot.data.k27.internal.gene.body.1,
        MARGIN = 1,
        FUN = max, na.rm = T)
max.metagenepplot.data.k27.internal.gene.body.2 <-
  apply(X = metagenepplot.data.k27.internal.gene.body.2,
        MARGIN = 1,
        FUN = max, na.rm = T)
max.metagenepplot.data.k27.internal.gene.body.3 <-
  apply(X = metagenepplot.data.k27.internal.gene.body.3,
        MARGIN = 1,
        FUN = max, na.rm = T)

max.metagenepplot.data.k27.internal.gene.body <- (max.metagenepplot.data.k27.internal.gene.body.1 +
                                                    max.metagenepplot.data.k27.internal.gene.body.2 +
                                                    max.metagenepplot.data.k27.internal.gene.body.3)/3

max.metagenepplot.data.k27.tss.1 <-
  apply(X = metagenepplot.data.k27.tss.1,
        MARGIN = 1,
        FUN = max, na.rm = T)
max.metagenepplot.data.k27.tss.2 <-
  apply(X = metagenepplot.data.k27.tss.2,
        MARGIN = 1,
        FUN = max, na.rm = T)
max.metagenepplot.data.k27.tss.3 <-
  apply(X = metagenepplot.data.k27.tss.3,
        MARGIN = 1,
        FUN = max, na.rm = T)
max.metagenepplot.data.k27.tss <- (max.metagenepplot.data.k27.tss.1 +
                                   max.metagenepplot.data.k27.tss.2 +
                                   max.metagenepplot.data.k27.tss.3)/3
```

```

max.metagenepplot.data.k27.tes.1 <-
  apply(X = metagenepplot.data.k27.tes.1,
        MARGIN = 1,
        FUN = max, na.rm = T)
max.metagenepplot.data.k27.tes.2 <-
  apply(X = metagenepplot.data.k27.tes.2,
        MARGIN = 1,
        FUN = max, na.rm = T)
max.metagenepplot.data.k27.tes.3 <-
  apply(X = metagenepplot.data.k27.tes.3,
        MARGIN = 1,
        FUN = max, na.rm = T)
max.metagenepplot.data.k27.tes <- (max.metagenepplot.data.k27.tes.1 +
  max.metagenepplot.data.k27.tes.2 +
  max.metagenepplot.data.k27.tes.3)/3

shapiro.test(max.metagenepplot.data.k27.gene.body)

##
##  Shapiro-Wilk normality test
##
## data:  max.metagenepplot.data.k27.gene.body
## W = 0.85035, p-value = 1.733e-11

shapiro.test(max.metagenepplot.data.k27.internal.gene.body)

##
##  Shapiro-Wilk normality test
##
## data:  max.metagenepplot.data.k27.internal.gene.body
## W = 0.554, p-value < 2.2e-16

shapiro.test(max.metagenepplot.data.k27.tss)

##
##  Shapiro-Wilk normality test
##
## data:  max.metagenepplot.data.k27.tss
## W = 0.92841, p-value = 0.003157

shapiro.test(max.metagenepplot.data.k27.tes)

##
##  Shapiro-Wilk normality test
##
## data:  max.metagenepplot.data.k27.tes
## W = 0.91454, p-value = 9.955e-05

wilcox.test(x = max.metagenepplot.data.k27.gene.body,
            y = max.metagenepplot.data.k27.internal.gene.body,
            alternative = "greater")$p.value

## [1] 1.317676e-42

wilcox.test(x = max.metagenepplot.data.k27.gene.body,
            y = max.metagenepplot.data.k27.tss,
            alternative = "greater")

```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data: max.metagenesplot.data.k27.gene.body and max.metagenesplot.data.k27.tss
## W = 6403, p-value = 6.017e-08
## alternative hypothesis: true location shift is greater than 0
```

```
wilcox.test(x = max.metagenesplot.data.k27.gene.body,
            y = max.metagenesplot.data.k27.tss,
            alternative = "greater")
```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data: max.metagenesplot.data.k27.gene.body and max.metagenesplot.data.k27.tss
## W = 9627, p-value = 6.942e-15
## alternative hypothesis: true location shift is greater than 0
```

```
wilcox.test(x = max.metagenesplot.data.k27.tss,
            y = max.metagenesplot.data.k27.tss)
```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data: max.metagenesplot.data.k27.tss and max.metagenesplot.data.k27.tss
## W = 2290, p-value = 0.1596
## alternative hypothesis: true location shift is not equal to 0
```

```
wilcox.test(x = max.metagenesplot.data.k27.tss,
            y = max.metagenesplot.data.k27.tss)
```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data: max.metagenesplot.data.k27.tss and max.metagenesplot.data.k27.tss
## W = 2290, p-value = 0.1596
## alternative hypothesis: true location shift is not equal to 0
```

Indeed the differences between the metagenesplots are highly significant except the ones corresponding to the marks overlapping with the TSS or TES.

Now we analyse the gene expression level of marked genes.

```
temp.gene.expression <- read.table(file="rnaseq_data/ostta_temperature_gene_expression.tsv",header = T,
gene.expression.20 <-
  (temp.gene.expression$T20_1 +
   temp.gene.expression$T20_2 +
   temp.gene.expression$T20_3)/3

names(gene.expression.20) <- rownames(temp.gene.expression)

non.k27.genes <- setdiff(names(gene.expression.20),c(k27.gene.body,k27.tss,k27.tss,k27.internal.gene.bo

summary(gene.expression.20[k27.gene.body])
```

```
##      Min.    1st Qu.    Median      Mean    3rd Qu.      Max.
```

```
## 0.00425 0.19081 0.90669 12.61720 4.35649 184.25682
```

```
summary(gene.expression.20[k27.internal.gene.body])
```

```
##      Min.   1st Qu.   Median     Mean  3rd Qu.    Max.
## 0.0042  6.1744  14.2076  54.7586  35.0676  785.5262
```

```
summary(gene.expression.20[k27.tss])
```

```
##      Min.   1st Qu.   Median     Mean  3rd Qu.    Max.
## 0.0042  0.4393  2.9986  33.4461  14.0569  566.7411
```

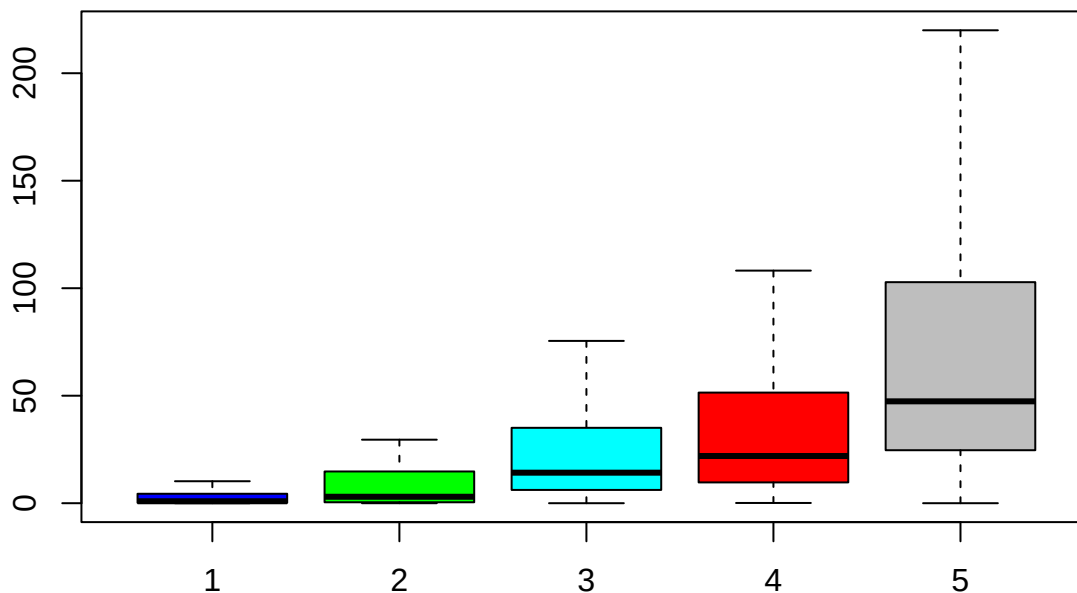
```
summary(gene.expression.20[k27.tes])
```

```
##      Min.   1st Qu.   Median     Mean  3rd Qu.    Max.
## 0.1178  9.8206  21.9533  47.6954  51.0203  297.9537
```

```
summary(gene.expression.20[non.k27.genes])
```

```
##      Min.   1st Qu.   Median     Mean  3rd Qu.    Max.
## 0.004  24.641  47.380  124.142  102.811  13604.318
```

```
#jpeg(filename = "images/h3k27me3_expression_barplot_different_genes.jpg",res = 250,width = 600,height = 400)
boxplot(gene.expression.20[k27.gene.body],
        gene.expression.20[k27.tss],
        gene.expression.20[k27.internal.gene.body],
        gene.expression.20[k27.tes],
        gene.expression.20[non.k27.genes],
        col=c("blue","green","cyan","red","grey"),
        outline=F)
```



```
#dev.off()
```

```
wilcox.test(x = gene.expression.20[k27.gene.body],
            y = gene.expression.20[k27.tss],
            alternative = "less")
```

```
##
## Wilcoxon rank sum test with continuity correction
```

```
##
## data: gene.expression.20[k27.gene.body] and gene.expression.20[k27.tss]
## W = 3352.5, p-value = 0.006989
## alternative hypothesis: true location shift is less than 0
wilcox.test(x = gene.expression.20[k27.tss],
            y = gene.expression.20[k27.internal.gene.body],
            alternative = "less")

##
## Wilcoxon rank sum test with continuity correction
##
## data: gene.expression.20[k27.tss] and gene.expression.20[k27.internal.gene.body]
## W = 3109.5, p-value = 5.199e-07
## alternative hypothesis: true location shift is less than 0
wilcox.test(x = gene.expression.20[k27.internal.gene.body],
            y = gene.expression.20[k27.tss],
            alternative = "less")

##
## Wilcoxon rank sum test with continuity correction
##
## data: gene.expression.20[k27.internal.gene.body] and gene.expression.20[k27.tss]
## W = 6295, p-value = 0.01968
## alternative hypothesis: true location shift is less than 0
wilcox.test(x = gene.expression.20[k27.tss],
            y = gene.expression.20[non.k27.genes],
            alternative = "less")

##
## Wilcoxon rank sum test with continuity correction
##
## data: gene.expression.20[k27.tss] and gene.expression.20[non.k27.genes]
## W = 170128, p-value = 3.801e-08
## alternative hypothesis: true location shift is less than 0
```

We can see all H3K27me3 genes present levels of expression significantly lower than non-marked genes. Specifically, genes whose entire gene body or TSS is occupied by H3K27me3 are completely repressed. Whereas, genes where only the TES or gene body is occupied by H3K27me3 are expressed although at a significantly lower level than non-marked genes.

In what follows we perform a functional analysis over the H3K27me3 marked genes. First, we analyze the relationship between H3K27me3 peaks and transposable elements.

```
dna.transposons <- read.table(file="te_analysis/dna_transposons_unsorted.bed", sep="\t", header=F)
nrow(dna.transposons)
```

```
## [1] 35
```

```
ltr.transposons <- read.table(file="te_analysis/ltr_transposons_unsorted.bed", sep="\t", header=F)
nrow(ltr.transposons)
```

```
## [1] 69
```

```
unclassified.transposons <- read.table(file="te_analysis/unknown_transposons_unsorted.bed", sep="\t", header=F)
nrow(unclassified.transposons)
```

```
## [1] 988
```

```

overlapping.peaks <- function(target.peak, peaks)
{
  subset(peaks, V1 == target.peak[1] & (
    (V2 >= target.peak[2] & V2 <= target.peak[3]) |
    (V3 >= target.peak[2] & V3 <= target.peak[3]) |
    (V2 <= target.peak[2] & V3 >= target.peak[3])))
}

peak.te <- vector(mode = "character", length = nrow(ll.20.peak.annotation))

for(i in 1:nrow(ll.20.peak.annotation))
{
  current.peak <- unlist(ll.20.peak.annotation[i,1:3])

  if(nrow(overlapping.peaks(current.peak,dna.transposons)) > 0)
  {
    peak.te[i] <- "DNA_transposon"
  } else if (nrow(overlapping.peaks(current.peak,ltr.transposons)) > 0)
  {
    peak.te[i] <- "LTR_transposon"
  } else if (nrow(overlapping.peaks(current.peak,unclassified.transposons)) > 0)
  {
    peak.te[i] <- "Unclassified_transposon"
  } else
  {
    peak.te[i] <- "No_transposon"
  }
}

100*table(peak.te)/sum(table(peak.te))

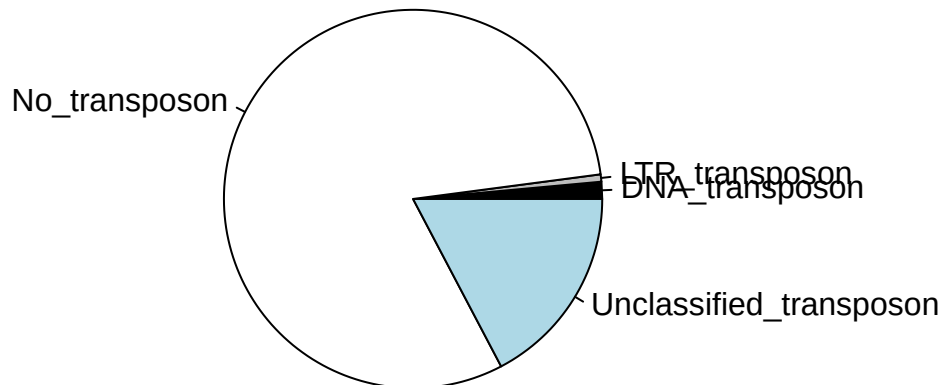
```

```

## peak.te
##          DNA_transposon          LTR_transposon          No_transposon
##             1.458333              0.625000             80.625000
## Unclassified_transposon
##             17.291667

pie(table(peak.te), col=c("black","grey","white","lightblue"))

```



```
nrow(dna.transposons)
```

```

## [1] 35
100*table(peak.te)["DNA_transposon"]/nrow(dna.transposons)

## DNA_transposon
##          20
nrow(ltr.transposons)

## [1] 69
100*table(peak.te)["LTR_transposon"]/nrow(ltr.transposons)

## LTR_transposon
##      4.347826
nrow(unclassified.transposons)

## [1] 988
100*table(peak.te)["Unclassified_transposon"]/nrow(unclassified.transposons)

## Unclassified_transposon
##          8.40081
table(ll.20.peak.annotation$peak.annotation[peak.te == "DNA_transposon"])

##
##      gene.body      intergenic internal.gene.body
##          2          4          1
table(ll.20.peak.annotation$peak.annotation[peak.te == "LTR_transposon"])

##
## gene.body intergenic
##      1      2
table(ll.20.peak.annotation$peak.annotation[peak.te == "Unclassified_transposon"])

##
##      gene.body      intergenic internal.gene.body      TES
##          24          25          14          13
##      TSS
##          7
table(ll.20.peak.annotation$peak.annotation)

##
##      gene.body      intergenic internal.gene.body      TES
##          114          38          201          74
##      TSS
##          53
(table(ll.20.peak.annotation$peak.annotation[peak.te == "DNA_transposon"])["intergenic"] +
table(ll.20.peak.annotation$peak.annotation[peak.te == "LTR_transposon"])["intergenic"] +
table(ll.20.peak.annotation$peak.annotation[peak.te == "Unclassified_transposon"])["intergenic"]) / tab

## intergenic
## 0.8157895
ll.20.peak.annotation <- cbind(ll.20.peak.annotation,peak.te)
write.table(x = ll.20.peak.annotation,file = "tables/H3K27me3_LL_20C_peak_annotation.tsv",quote = F,sep

```

H3K27me3 is only partially associated with Transposable Elements. More than 81% of the peaks did not overlap with any TE. Only 1.5%, 0.6% and 17.3% of the H3K27me3 peaks were found overlapping with DNA, LTR and unclassified transposons respectively. Most TE were neither associated with H3K27me3 peaks, specifically 7 out of 35 (20%) DNA transposons, 3 out of 69 (4.3%) LTR transposons and 83 out of 988 unclassified transposons were H3K27me3 marked. Most H3K27me3 marks located at intergenic regions (81.6%) were associated with transposons